Mohit Marvania

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RESEARCH INTERESTS

LLM alignment robustness – specifically whether safety and capability properties survive adversarial stress-testing, sequential fine-tuning, and distribution shift. Interests span reinforcement learning from human feedback, empirical AI safety evaluations, and scalable oversight.

EDUCATION

George Mason University — **Fairfax, Virginia**
M.S. in Computer Science – Machine Learning — *Expected, May 2026*

Charotar University of Science & Technology — **Changa, India**
Bachelor of Technology in Computer Science (GPA: 3.78/4.0) — *Graduated, April 2024*

RESEARCH EXPERIENCE & PUBLICATIONS

Graduate Research Assistant — Fairfax, VA
Supervisor: Dr. Abolfazl Safikhani & Dr. Ziwei Zhu, George Mason University. — *Jan 2025 – Present*

Catastrophic Forgetting in Continual Preference Alignment of LLMs
Co-First Author | Advisor: Dr. Ziwei Zhu — *Targeting (ARR May 2026)*
- Found that safety alignment degrades silently during sequential fine-tuning, and that the degradation pattern depends on task ordering and algorithm choice more than model size. First systematic comparison of PPO, DPO, and GRPO across 5 alignment dimensions and 11 task orderings.
- Owned the complete GRPO implementation end-to-end, reward model training, policy fine-tuning on Llama-3.1-8B and Qwen3-4B, BWT metrics evaluation across all sequence-model combinations on Slurm-managed GPU clusters.

Toward Resilient Watermark Detection: Stability-Aware Statistical Features
Co-First Author | Advisor: Dr. Abolfazl Safikhani — *Under Review (ICML 2026)*
- Identified a critical failure mode in LLM watermarking: standard detectors collapse below 60% AUC under recursive paraphrasing, a real-world attack any adversary can run. Built PSS framework that holds at 94.6% AUC at depth-6 and greater than 81% AUC on 300-token texts where prior methods failed entirely.
- Led paper writing and co-designed the adversarial evaluation pipeline across Mistral-7B, Llama-2, Gemma, and Qwen, optimized detection to O(n) complexity via rolling window analysis on A100s.

VeriSim: A Configurable Framework for Evaluating Medical AI Under Realistic Patient Noise
Co-First Author | Advisor: Dr. Ziwei Zhu — *Targeting (ARR May 2026)*
- Showed that SOTA medical LLMs lose 15-25% diagnostic accuracy when patient simulators behave like real patients, not idealized benchmark inputs. Current static benchmarks are measuring something different from real-world performance.
- Designed the patient simulator, 6-pillar clinical noise injection framework, and LLM-as-a-Judge along with human evaluation pipeline (Cohen's Kappa=0.81 human agreement).

Reasoning Under Constraint: How Batch Prompting Suppresses Overthinking
Contributor | Independent Collaboration — *Accepted (ICLR 2026 Workshop on LLM Reasoning)*
- Batch prompting cuts reasoning tokens 76% while preserving or improving accuracy across 13 benchmarks, an inference-time technique that requires no model modification, applicable to closed APIs like OpenAI-o1.

Machine Learning Researcher Intern — Changa, India
Supervisor: Dr. Amit Thakkar, Charusat University. — *Dec 2023 – Apr 2024*

YOLOv8-based Emotion Recognition for Student Engagement Assessment
First Author — *Accepted (AIP Conference Proceedings, 2025)*
- Built and published real-time engagement detection system achieving 72.3% accuracy across 6 behavioral classes, automated dataset fusion pipeline (4000+ samples) cutting preprocessing time by 80%.

INDUSTRY EXPERIENCE

Software Engineer Intern — Rajkot, India
Raven Technolabs — *May 2023 – Jul 2023*
- Reduced database write latency 30% for AI-assisted receipt system serving 2k+ weekly users via batched operations, introduced CI/CD pipelines that cut post-release hotfixes.

SKILLS

Languages: Python (Advanced), C++, Java, JavaScript/TypeScript, Bash, SQL
RLHF & Alignment: PPO, DPO, GRPO, reward modeling, preference optimization, continual alignment, scalable oversight
ML & Safety Stack: PyTorch, HuggingFace, Transformers, TRL, DeepSpeed, Unsloth, Weights & Biases
Models: Llama-3, Mistral, Qwen, DeepSeek-R1, GPT-4o API, Claude API
Infrastructure: Slurm (HPC cluster management), Docker, AWS (EC2/S3), Linux (Ubuntu)